Maeve Hutchinson

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Education

PhD in Computer Science — City St George's, University of London
Thesis: Data Visualisation Mediated through Natural Language Interaction — 02/2023 – 03/2027 *(Anticipated)*

- Committee: Dr Pranava Madhyastha, Dr Radu Jianu, and Dr Aidan Slingsby.
- Awarded a 3-year Doctoral Studentship by the School of Science and Technology.
- Researching the use of LLMs and Multimodal LLMs to aid users in visual analytic systems.
- Building real-world datasets and benchmarks to test and improve models' visual reasoning capabilites.

MSc in Data Science *(with Distinction)* — City St George's, University of London
Thesis: Extraction and Evaluation of Causal Relations From Aero-Engineering Text — 09/2021 – 01/2023

- Used BERT to identify fault-related causal relationships from unstructured technical documentation to support diagnostic and maintenance processes in aero-engineering.

BSc in Physics — Imperial College London
Dissertation: Agent-Based Modelling as a Tool for Understanding Financial Crises — 12/2018 – 06/2021

- Explored the use of physics-based models from information theory as applied to model finanical markets.

Selected Publications

Journal Papers:

Maeve Hutchinson, Radu Jianu, Aidan Slingsby & Pranava Madhyastha. *Foundation Model Assisted Visual Analytics: Opportunities and Challenges.*, Computers & Graphics, vol. 130, p. 104246, Aug. 2025. doi:10.1016/j.cag.2025.104246

Conference Papers:

Maeve Hutchinson, Syed Mahbubul Huq, Mohammad Albinhassan, Radu Jianu, Aidan Slingsby & Pranava Madhyastha. *InSight: A Benchmark for Agentic Claim Verification in Interactive Visualizations* Empirical Methods in Natural Language Processing (EMNLP), Oct. 2026. doi:10.48550/arXiv.2609.01383

Maeve Hutchinson, Abderrahmane Wassim Mehdaoui & Pranava Madhyastha. *Attention-Guided Saliency Maps for Interpreting Visualization Literacy in VLMs.* IEEE Visualization and Visual Analytics (VIS), Nov. 2026. doi:10.48550/arXiv.2607.16105

Maeve Hutchinson, Radu Jianu, Aidan Slingsby, Jo Wood & Pranava Madhyastha. *Capturing Visualization Design Rationale.* IEEE Visualization and Visual Analytics (VIS), Nov. 2025. doi:10.1109/VIS60296.2025.00052

Radu Jianu, Maeve Hutchinson Natalia Andrienko, Gennady Andrienko, Mai Elshehaly, & Aidan Slingsby. *Mind-Mapping Data Analysis with LLMs: From Vision to First Steps.* Computer Graphics & Visual Computing (CGVC), Sep. 2025. doi:10.2312/cgvc.20251221

Maeve Hutchinson, Radu Jianu, Aidan Slingsby, Jo Wood, & Pranava Madhyastha. *Chart Question Answering from Real-World Analytical Narratives.* Association for Computational Linguistics (ACL), Jul. 2025. doi:10.18653/v1/2025.acl-srw.50

Maeve Hutchinson, Radu Jianu, Aidan Slingsby & Pranava Madhyastha. *LLM-Assisted Visual Analytics: Opportunities and Challenges.* In: Computer Graphics & Visual Computing (CGVC), Sep. 2024. doi:10.2312/cgvc.20241237

Maeve Hutchinson, Aidan Slingsby, Radu Jianu & Pranava Madhyastha. *Towards Visualisation Specifications from Multilingual Natural Language Queries using Large Language Models.*. EuroVis, Jun. 2023. doi:10.2312/evp.20231072

Awards and Prizes

Best Full Paper Award Computer Graphics & Visual Computing 2024
EurAI Financed Grant IJCAI 2026

Teaching Experience

Graduate Teaching Assistant City St George's, University of London
Neural Computing; Big Data; Data Visualization; Mathematics for Computing 10/2023 – *present*

Associate Fellow, Advance Higher Education (AFHEA) 10/2024

Academic Service

Reviewer ACL Rolling Review, ACL FEVER, Computers & Graphics Journal, EuroVis, IEEE Visualization and Visual Analytics

Programme Committee IEEE VIS Posters, Computer Graphics & Visual Computing

Student Volunteer IJCAI 2026, EMNLP 2026